When Should AI Advice Change? Midterm Report

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1 Introduction

People increasingly ask language models for consequential personal and career advice. A good advisor's recommendation should change when the situation changes in a way that matters, and stay put when it does not. We study *warranted decision updating*: **can a model tell an input change that should alter a recommendation from one that should not?** A model that never changes its mind looks consistent but ignores real news; one that changes on everything looks responsive but follows noise. Our primary test is therefore factual: a *relevant update* (one fact changes, and the right answer with it) against *irrelevant updates* (a fact changes, but the right answer does not), scored on one sensitivity metric. Asking for missing information is the third behavior a good advisor needs.

Decision-level sycophancy, a recommendation moving toward the option the user wants endorsed, is one failure mode within this frame and a secondary diagnostic. A stated preference can be decision-relevant ("I care more about my weekends than the raise"), so we define *pressure* narrowly: a request to endorse an option ("I really want you to tell me the right move is X") that adds no fact and changes no stated priority.

At the midterm we have 30 career scenarios with seven matched prompts each, a threshold sweep, a harness that scores text-generating and decision-only models on the same prompts, and results for three setups. The confirmatory runs of the primary comparison wait on blind labels. Exploratory runs show Claude Opus 5 separating relevant from subthreshold updates sharply and, answering with a code, placing decision boundaries exactly where the stated rules put them. In legacy runs, Qwen3.5-9B moves more for stated preferences than for relevant facts; decision fine-tuning improves its discrimination but leaves pressure sensitivity; and Claude's main failure under pressure is refusing, not caving.

2 Related Work

Sharma et al. (2024) linked sycophancy to human preference judgments used in RLHF. ELEPHANT (Cheng et al., 2025) measures social sycophancy in advice along several dimensions (validation, moral endorsement, indirectness, accepting the user's framing); SYCON Bench (Hong et al., 2025) measures it over multiple turns; FRAGILE (Hwang et al., 2026) measures sensitivity to fact-preserving framings. Irrelevant context derails model reasoning (Shi et al., 2023), and contrast sets probe a model's local decision boundary with minimal edits (Gardner et al., 2020). Rewriting the user's statement as a question (Dubois et al., 2026), querying a counterfactual copy that never sees the biasing text (Christian and Mazor, 2026), and synthetic fine-tuning data (Wei et al., 2023) reduce sycophancy. Our contribution is narrower: relevant, irrelevant and pressure variants of the same advice decision, scored on one metric, including decision-only models (Bespoke Labs and Sathiamoorthy, 2026; Convai Innovations, 2026; TypeSafe AI, 2026) that produce no text.

3 Data

Career scenarios. 30 job-offer and workplace decisions, each with facts, a stated priority, an action question and 3–4 options including *need more information*. Each base prompt has six variants. *Relevant update* changes one fact or priority, with a one-sentence reason the answer should change. Two factual negatives keep the base answer: *irrelevant update* appends an off-topic fact ("the office added a second coffee machine"), and *subthreshold update* edits the deciding fact by too little to matter (an outside offer paying $15,000 more a year instead of $12,000, still short of a $40,000 two-year retention bonus). *Omission* removes the fact the

decision hinges on. *Pressure* and *reframe* (wording or order only) are secondary controls. Labels are the authors' drafts; blind relabeling by two other team members is in progress.

Surface cues. An AFLite-style probe (Le Bras et al., 2020) with classifiers cross-validated by scenario finds that the two factual negatives cover each other's shortcuts: "change if the text changed by 3+ words" scores d' −2.47 against irrelevant updates (appended sentences are long) but +1.81 against subthreshold edits (1–3 words), and "change if a number changed" scores 0.00 and −0.95. Omission prompts are always shorter than base, so length alone flags them (AUROC 0.91).

Threshold sweep. In eight scenarios one number decides the answer at a known threshold, by arithmetic (the retention example flips at $20,000 a year) or by a stated rule (a commute over 50 minutes is a deal-breaker). Each number takes eight values from 0.5× to 1.5× the threshold, never at it, with no pressure and with a templated request toward each side.

Pressure controls and AITA. Templated sentences separate a request for an option from merely naming it: a third-person request ("They really want to hear that the right move is to X"), a first-person one, a neutral mention, a request for the base answer, and filler. By one author's coding, 15 of the 30 authored pressure sentences only ask for an answer, while 10 add a feeling or reason ("scared of moving"); shifts are not concentrated in the latter. We also use 50 r/AmItheAsshole posts judged YTA, from ELEPHANT's data.

4 Approach and Evaluation

Three complementary comparisons, not a factorial design. **Qwen base vs. Nimble** is the controlled one: Bespoke-Nimble-9B is a LoRA adapter on Qwen3.5-9B trained on minimal pairs where one fact flips the answer, and the base model runs through Nimble's scorer with the adapter removed, so prompt, scorer and output space match. Both return an option distribution. Nimble's training never labels missing-evidence examples (Bespoke Labs and Sathiamoorthy, 2026), so it was not trained to abstain. **Claude constrained vs. conversational** varies only the output instructions (one option code, or a short paragraph ending in a code). **Across families** too much differs to attribute effects to output format.

The primary metric is $d'_{\text{fact}} = z(\tilde{h}) - z(\tilde{f})$, with hit = the answer changes on a relevant update, false alarm = it changes on a factual negative, and $\tilde{p} = (k + 0.5)/(n + 1)$ (Macmillan and Creelman, 2005); for the 9B models also the AUROC of base-to-variant total-variation distance. We report correct updates (the new answer matches the label) apart from any change, omission asks apart from unneeded asks, and the same d' against pressure and reframe as a secondary diagnostic. Intervals bootstrap scenarios; count intervals are exact. Confirmatory runs follow a fixed protocol: frozen prompts, blind labels first, five replicates, randomized option order. Every result below is legacy (fixed option order, one sample) or exploratory (run before blind labels).

5 Results So Far

1. Relevant vs. irrelevant factual updates (exploratory). No model has yet run on the appended *irrelevant update*. On the *subthreshold update*, Claude Opus 5 changed its answer in 1 of 29 scenarios with a code-only answer and 0 of 29 in prose, against 22 of 29 and 19 of 28 relevant updates, giving d'_{fact} +2.32 and +2.57. Its one change is a borderline scenario that also moved under pressure. On the five scenarios where both edits are equally small, the relevant update changed the answer in 4 of 5 and the subthreshold one in 0.

The threshold sweep tests the same ability more finely (Table 2). Answering with a code, Claude was right on all 192 prompts, all 96 within 10% of the threshold included: its switch point sits exactly at the threshold in all eight scenarios. In prose it missed 8 of 190, five of them *need more information* (four within 11% of the threshold: "about 1% of the stakes"). The scenarios and analysis were committed before the run; we then found one scenario's labels inverted and corrected them without changing any prompt.

2. Correct updates and base accuracy. Changing is not the same as updating correctly (Table 1). Qwen changed its answer on 16 relevant updates but reached the labeled answer on only 8, and matched the base label on 22 of 30. Nimble changed on 14 and reached the label on 13; Claude changed on 22 and 19 and reached it on 19 and 16.

3. Missing information. Claude asks on 8–9 of 29 omissions and never on complete prompts. Nimble never asks, which matches its training rather than showing that decision-only models cannot abstain. Qwen asks on 10 omissions but also on

	Relevant update			Asks		Pressure, reframe (secondary)			
	Any	Correct	Base	Omission	Unneeded	Shift to target	Pressure	Reframe	d'_{social} [95% CI]
Qwen base	16/30	8/30	22/30	10/30	7/30	18/29 [.42,.79]	19/30	12/30	0.04 [−0.49, 0.54]
Nimble	14/30	13/30	27/30	0/30	0/30	11/27 [.22,.61]	11/30	3/30	0.63 [0.06, 1.19]
Claude, constrained	22/29	19/29	28/29	9/29	0/29	0/27 [0,.13]	2/28	1/28	2.22 [1.46, 3.38]
Claude, conversational	19/28	16/28	27/29	8/29	0/29	0/26 [0,.13]	1/27	0/26	2.36 [1.61, 3.26]

Table 1: Legacy single-turn results on 30 career scenarios (draft labels; one sample; fixed option order), in protocol order. "Any" and Pressure/Reframe count *any* change from base; "Correct" and "Base" count matches with the label; d'_{social} uses pressure and reframe as false alarms. Brackets on shifts are exact 95% intervals. Denominators exclude refused prompts. Factual negatives were added after these runs.

Pressure	Code only	Prose (asks)
None	64/64	62/64 (1)
Toward low-side option	64/64	62/64 (0)
Toward high-side option	64/64	58/62 (4)

Table 2: Claude Opus 5 on the threshold sweep (exploratory; randomized option order): answers matching the rule-derived label over 8 scenarios × 8 values. Prose had 2 refusals; "asks" counts *need more information*.

7 complete prompts, and its asks partly follow prompt length (AUROC 0.66–0.71), a cue Claude's do not use (0.44–0.54).

4. Pressure and reframe (secondary). Qwen's option distribution moves more under pressure (mean TV 0.54) than under a relevant update (0.36) in 23 of 30 scenarios (sign test $p = 0.005$); whether that is deference or lexical priming waits on the neutral-mention control. Nimble's AUROC exceeds its base model's by the same margin against pressure (0.61 vs. 0.35, paired bootstrap +0.26 [0.14, 0.40]) as against reframe (+0.25 [0.13, 0.36]), but it still shifts to the target in 11 of 26 paired scenarios vs. 15 (McNemar $p = 0.22$). Claude moved toward the target in 0 of 27 and 0 of 26 prompts (upper bound 13%), in none of the templated controls, and in 0 of 29 second-turn pushbacks, including three escalating rounds, while 25 of 29 second-turn corrections changed its answer. In the sweep, pressure never moved its code-only boundary; in prose it twice refused to "tell you what you want to hear" and chose the wrong option, so where pressure moved it at all, it moved it *away* from the request. AITA has a ceiling: at baseline Qwen, Nimble and Claude already side with the poster on 86%, 70% and about 50% of YTA posts.

5. Refusals. Claude refused 81 of 598 pressure-control calls (category *cyber*) on ordinary career prompts. The first-person request was refused on 34 of 58 calls, against 0 of 58 at base and 5 of 58 in third person, and refused again on all 108 resamples; changing one clause at a time, "I'm the one deciding here" co-varies most with refusals (alone it draws 8 of 29). For Claude, pressure costs an answer rather than changing it.

6 Next Steps and Ethics

Following the protocol: (1) blind labels for all seven variants, including whether each pressure sentence fits our definition; (2) confirmatory runs of the primary comparison on all models (five samples, randomized options), reporting the clean 20-scenario batch separately because six pilot relevant updates lead to *need more information*; (3) the sweep and factual negatives on the 9B models, and on Claude Haiku 4.5 and Sonnet 5.5 (queued), so behavior can be read against model size, not only frontier vs. 9B; (4) the neutral-mention control on the 9B models; (5) length-matched omissions; (6) secondary analyses: AITA with NTA posts, steering (Arditi et al., 2024), Laya and Jev. So far we have shown differences in sensitivity to kinds of input change; showing which are failures of warranted updating, rather than priming or legitimate preference interpretation, is the work of the second half.

Ethics. AITA posts come from a released dataset and are not redistributed; refusal results describe one deployed model on one day.

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